

06 – Git & GitHub

A control version system and a server to store data

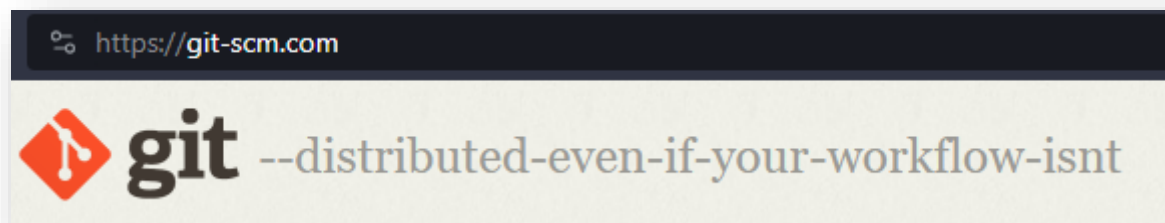
Jérôme Landré – jerome.landre@ju.se

OUTLINE

- 1) Git
- 2) GitHub
- 3) Step by Step
- 4) Add, Commit & Push
- 5) Others git commands

1) GIT

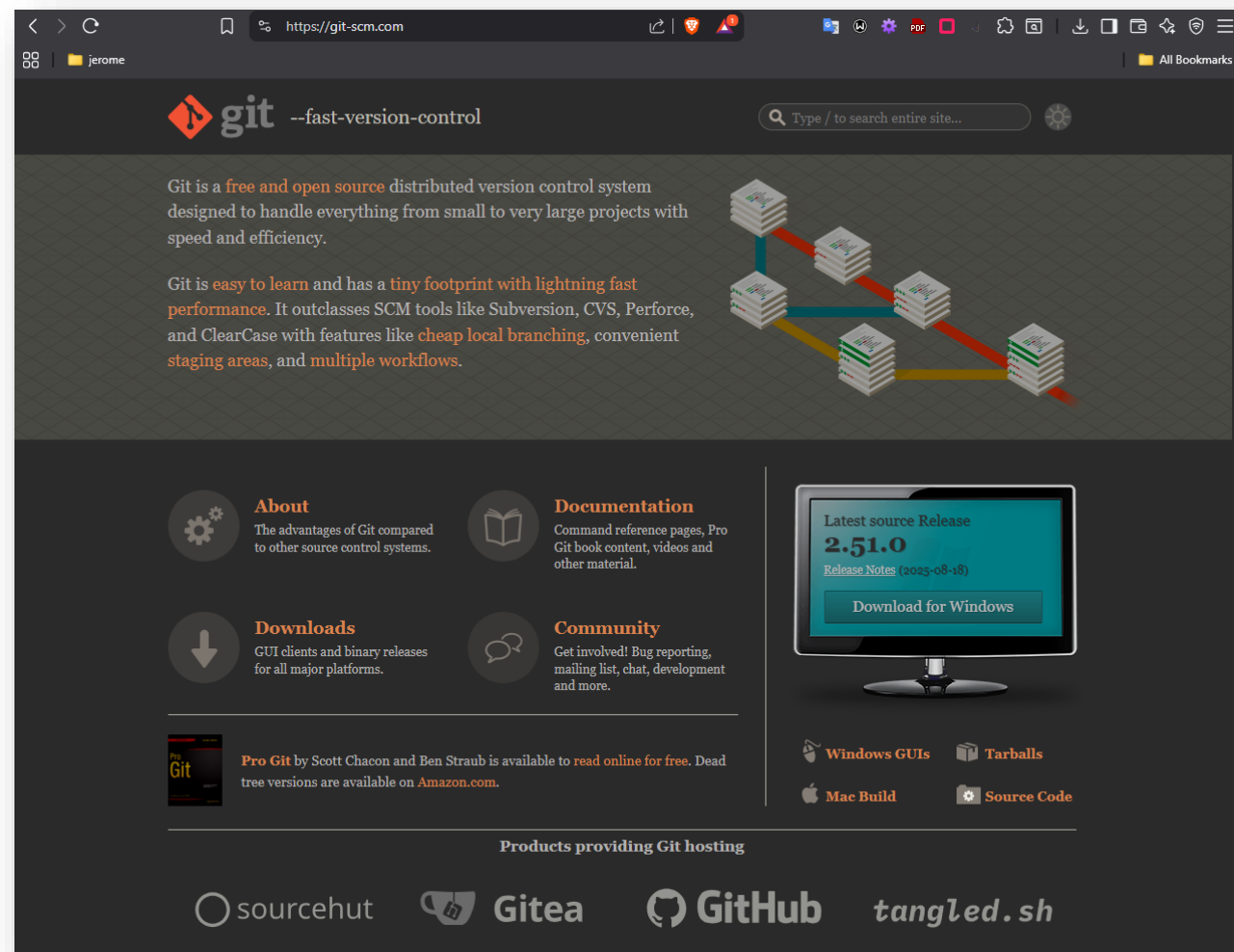
- “Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.
- Git is easy to learn and has a tiny footprint with lightning fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.” from <https://git-scm.com/>



1) GIT

- To be able to use git, you must install it on your system: Windows, MacOS, Linux.
- Download the installer and, install it
- Test: open a terminal and "git -v"

```
G:\>git -v  
git version 2.51.0.windows.1  
  
G:\>|
```

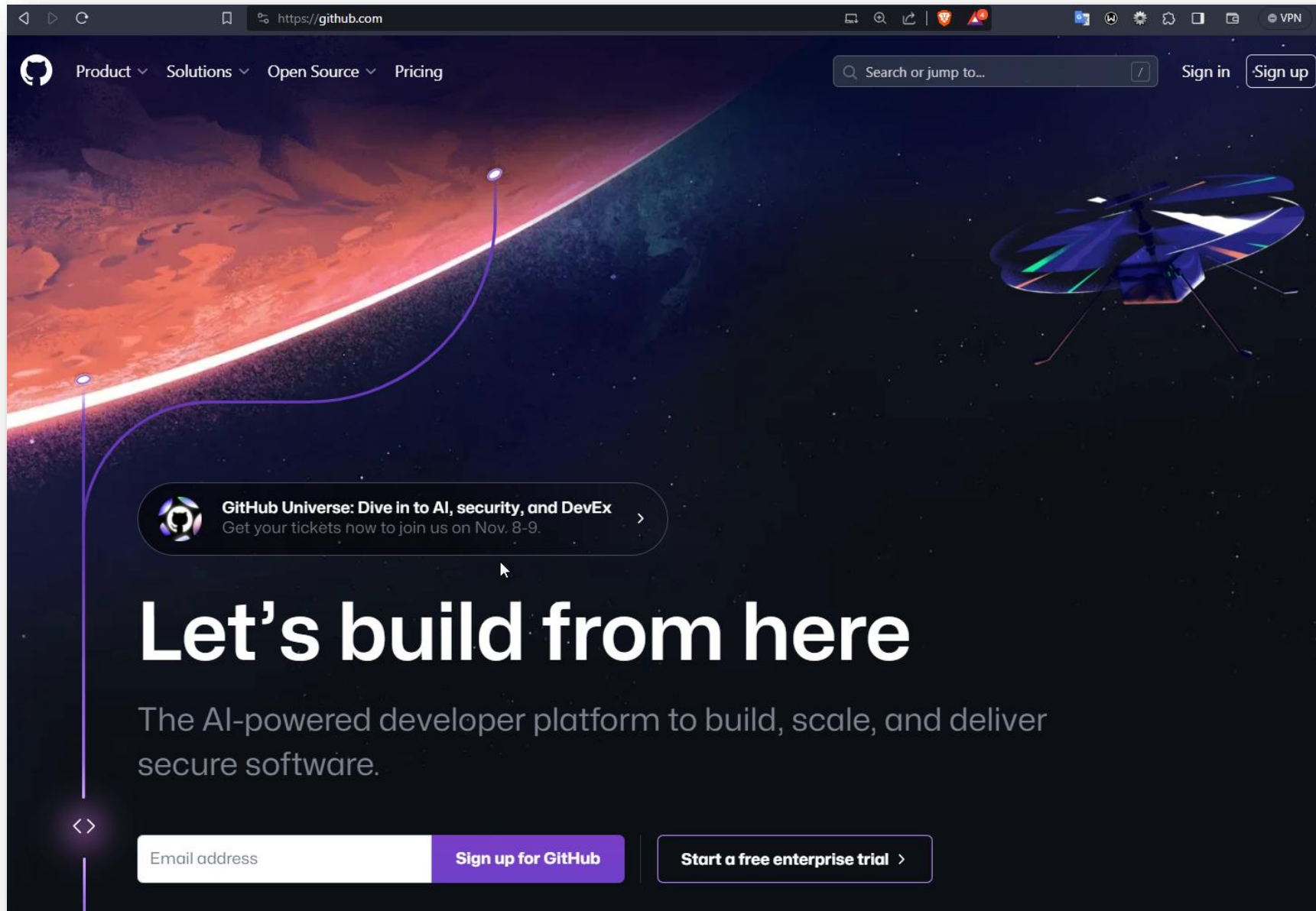


2) GITHUB

- **GitHub** is an Internet Server where you can store and share your codes online

<https://github.com/>

- **GitHub** understands natively **git** commands
- You need to create an account to use GitHub!



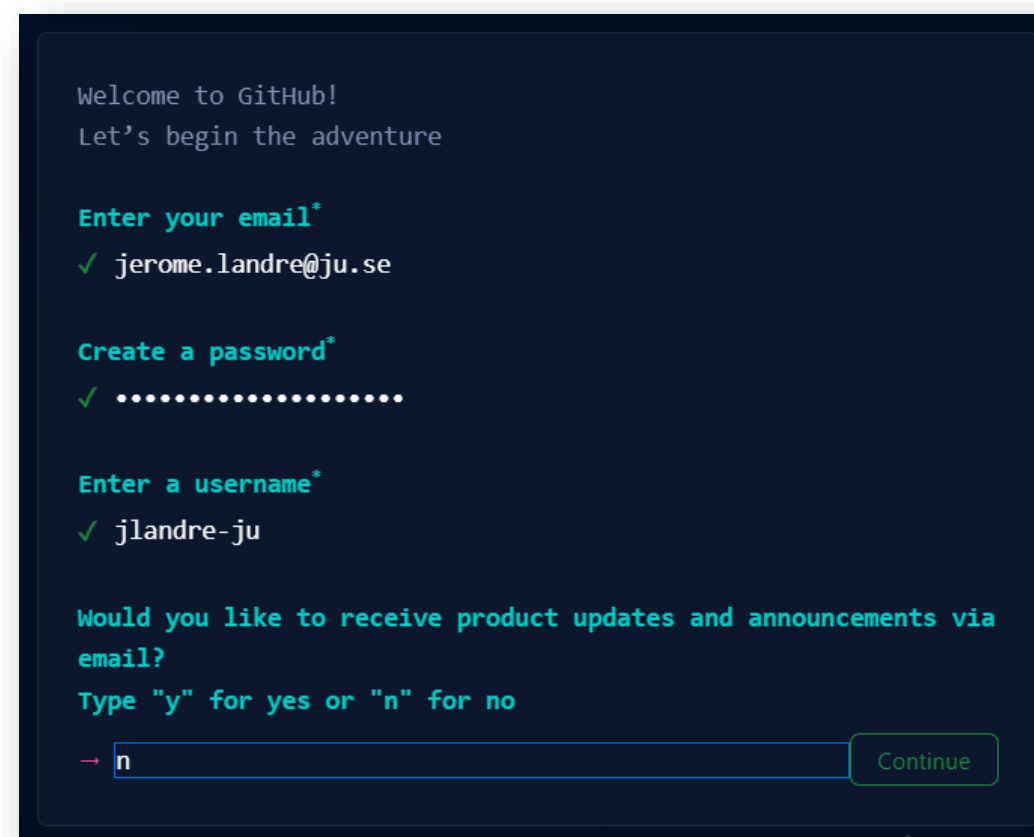
3) STEP BY STEP

1) Create an account on GitHub



A simulation of the GitHub sign-up form. It consists of a white rectangular input field on the left with the placeholder text "Email address", and a purple rectangular button on the right with the text "Sign up for GitHub". The entire form is set against a black background.

Remark: this is a **simulation**, I have already a GitHub account: **jlandre72!**



A simulation of the GitHub account creation process on a dark blue background. The text is in a light blue/cyan color. It starts with "Welcome to GitHub! Let's begin the adventure". Then it prompts "Enter your email*" with a green checkmark and the email "jerome.landre@ju.se". Next is "Create a password*" with a green checkmark and a series of dots. Then "Enter a username*" with a green checkmark and the username "jlandre-ju". Finally, it asks "Would you like to receive product updates and announcements via email?" with the instruction "Type 'y' for yes or 'n' for no". Below this is a text input field containing "n" and a green "Continue" button.

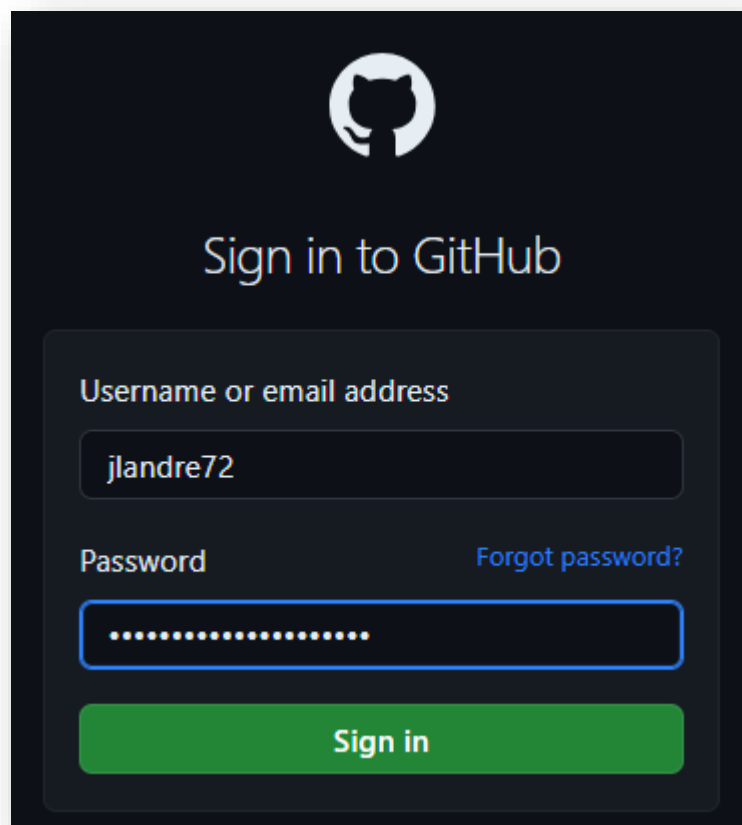
3) STEP BY STEP

2) Finish the setup of your account: confirm the registration by clicking on the link in the received email...

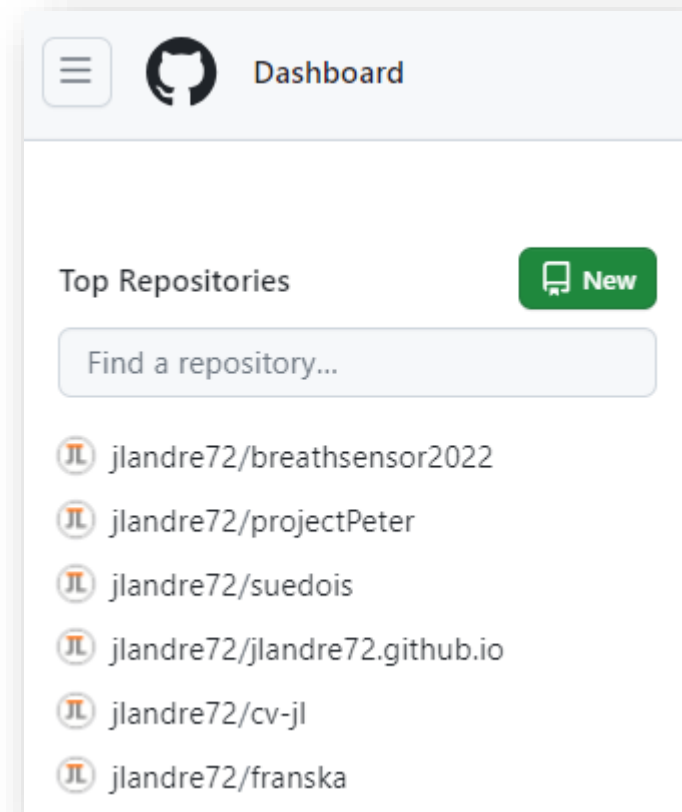
No picture because I have already an account so I cannot create a new one to the end...

3) STEP BY STEP

3) Log in on GitHub



The image shows the GitHub sign-in page. At the top is the GitHub logo. Below it is the text "Sign in to GitHub". There are two input fields: "Username or email address" with the text "jlandre72" and "Password" with masked characters. A link "Forgot password?" is next to the password field. At the bottom is a green "Sign in" button.



The image shows the GitHub dashboard for user jlandre72. At the top is the GitHub logo and the word "Dashboard". Below is a section "Top Repositories" with a green "New" button. There is a search bar "Find a repository...". Below the search bar is a list of repositories:

- jlandre72/breathsensor2022
- jlandre72/projectPeter
- jlandre72/suedois
- jlandre72/jlandre72.github.io
- jlandre72/cv-jl
- jlandre72/franska

3) STEP BY STEP

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk (*).

Repository template

No template ▾

Start your repository with a template repository's contents.

Owner *

 jlandre72 ▾

Repository name *

/ portfolio-jl

✔ portfolio-jl is available.

Great repository names are short and memorable. Need inspiration? How about [upgraded-succotash](#) ?

Description (optional)

My portfolio for the Web Dev Fun course! Written with Node, Express, Handlebars and SQLite3!

☐  Public

Anyone on the internet can see this repository. You choose who can commit.

☒  Private

You choose who can see and commit to this repository.

Initialize this repository with:

☒ Add a README file

This is where you can write a long description for your project. [Learn more about READMEs.](#)

4) Create a new repository: name it according to your type of project, it is a portfolio for me, so "portfolio-jl" makes sense!

Add .gitignore

.gitignore template: None ▾

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

Choose a license

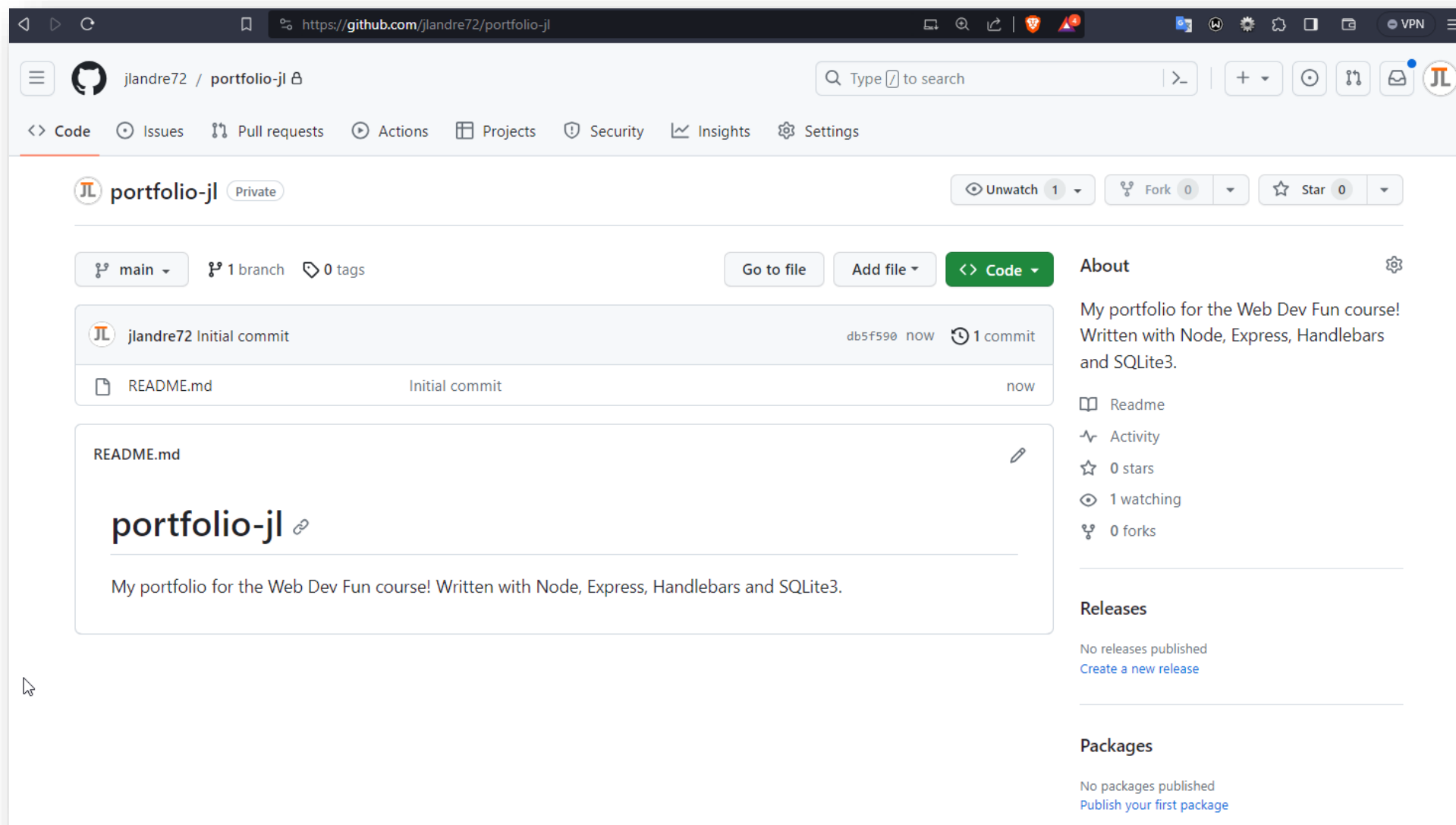
License: None ▾

A license tells others what they can and can't do with your code. [Learn more about licenses.](#)

This will set  main as the default branch. Change the default name in your [settings](#).

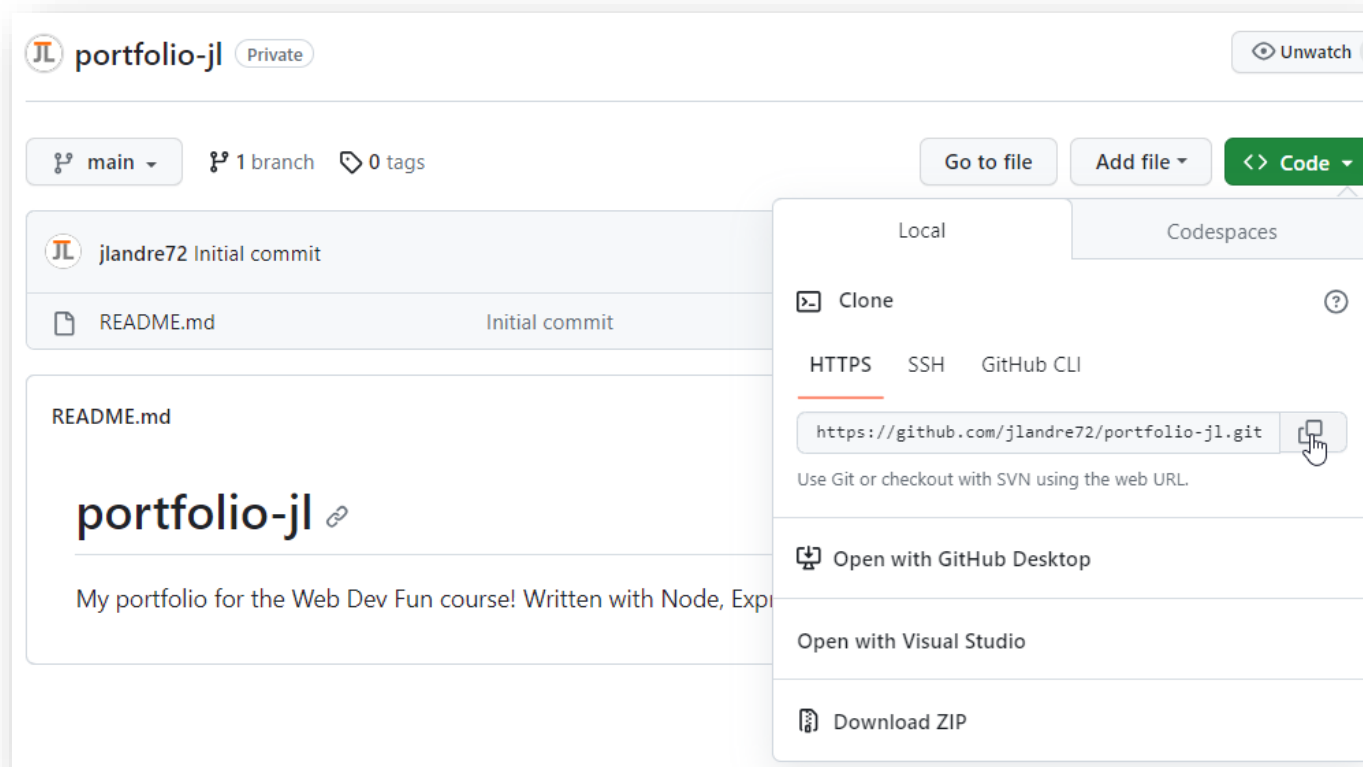
 You are creating a private repository in your personal account.

Create repository



Congratulations, you have created your GitHub repository! Now, the question is how to link it to your **project folder**?

- How to link it to your **project folder**?
- There are many ways to do so
- The simplest is to create a new folder that will synchronize to your GitHub repository
- On your GitHub repository page, go to the green "<> Code" button, click on it and copy the path with the double square button (see picture below)



- Open a terminal and go to a place you want to put your code: "cd Desktop"
- Create a new directory: "mkdir web-dev-fun"
- Go into your newly created directory: "cd web-dev-fun"
- Git clone your project: "git clone https://github.com/jlandre72/portfolio-jl.git"
- See the folder created: "dir"

git config --global user.name "jlandre72"

git config --global user.email "jerome.landre@ju.se"

- You have now your portfolio directory ready to synchronize with GitHub!

- Change the **information in red** with your own (of course!)

```
C:\Users\lanjer>cd Desktop

C:\Users\lanjer\Desktop>mkdir web-dev-fun

C:\Users\lanjer\Desktop>cd web-dev-fun

C:\Users\lanjer\Desktop\web-dev-fun>git clone https://github.com/jlandre72/portfolio-jl.git
Cloning into 'portfolio-jl'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (3/3), done.

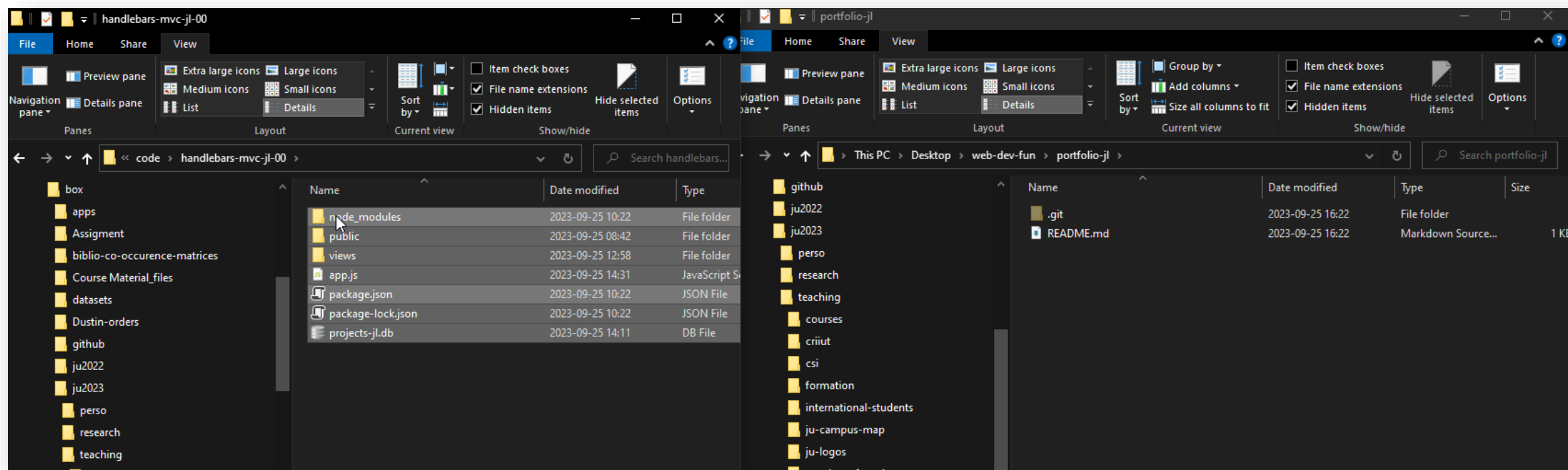
C:\Users\lanjer\Desktop\web-dev-fun>dir
Volume in drive C is Windows
Volume Serial Number is 2671-B1FB

Directory of C:\Users\lanjer\Desktop\web-dev-fun

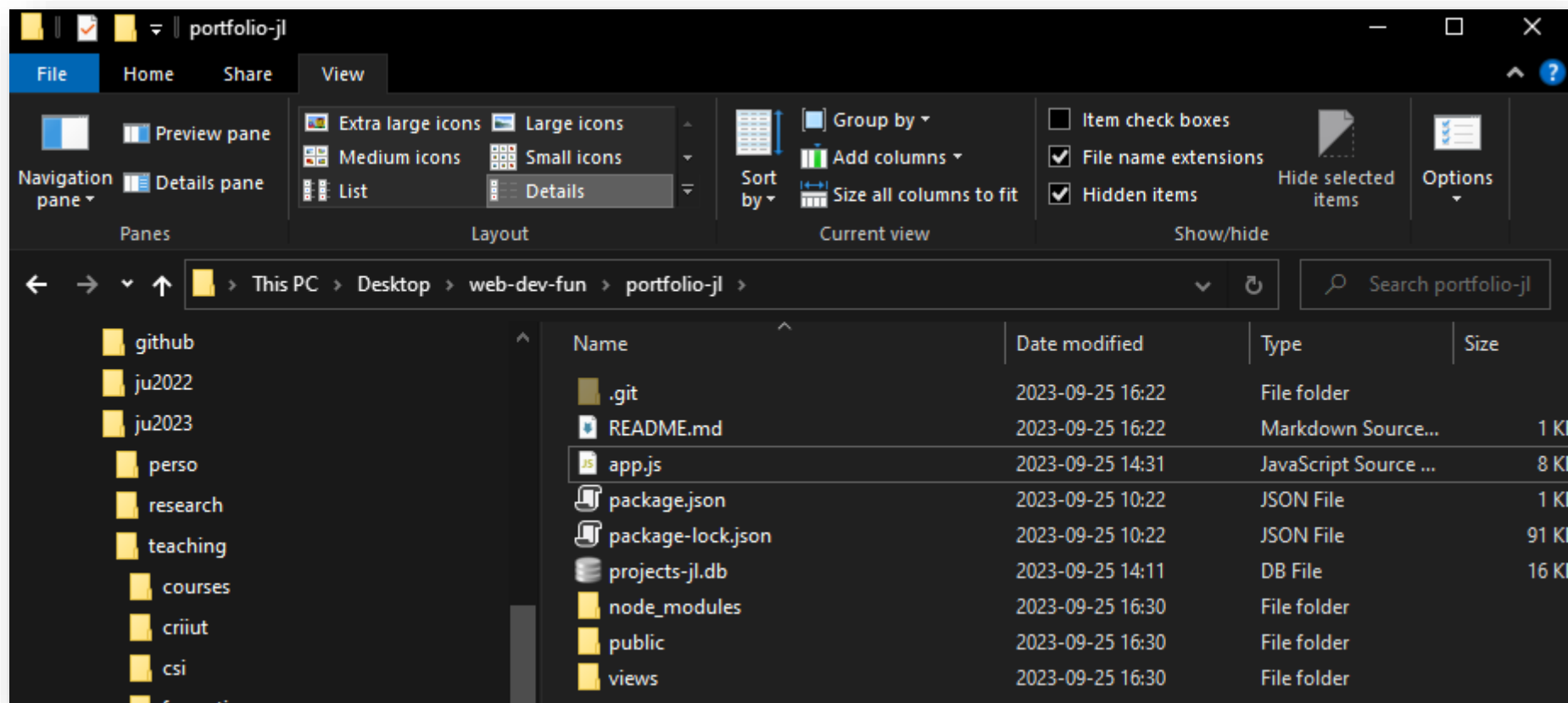
2023-09-25  16:22    <DIR>          .
2023-09-25  16:22    <DIR>          ..
2023-09-25  16:22    <DIR>          portfolio-jl
                   0 File(s)                0 bytes
                   3 Dir(s)  19 252 510 720 bytes free

C:\Users\lanjer\Desktop\web-dev-fun>
```

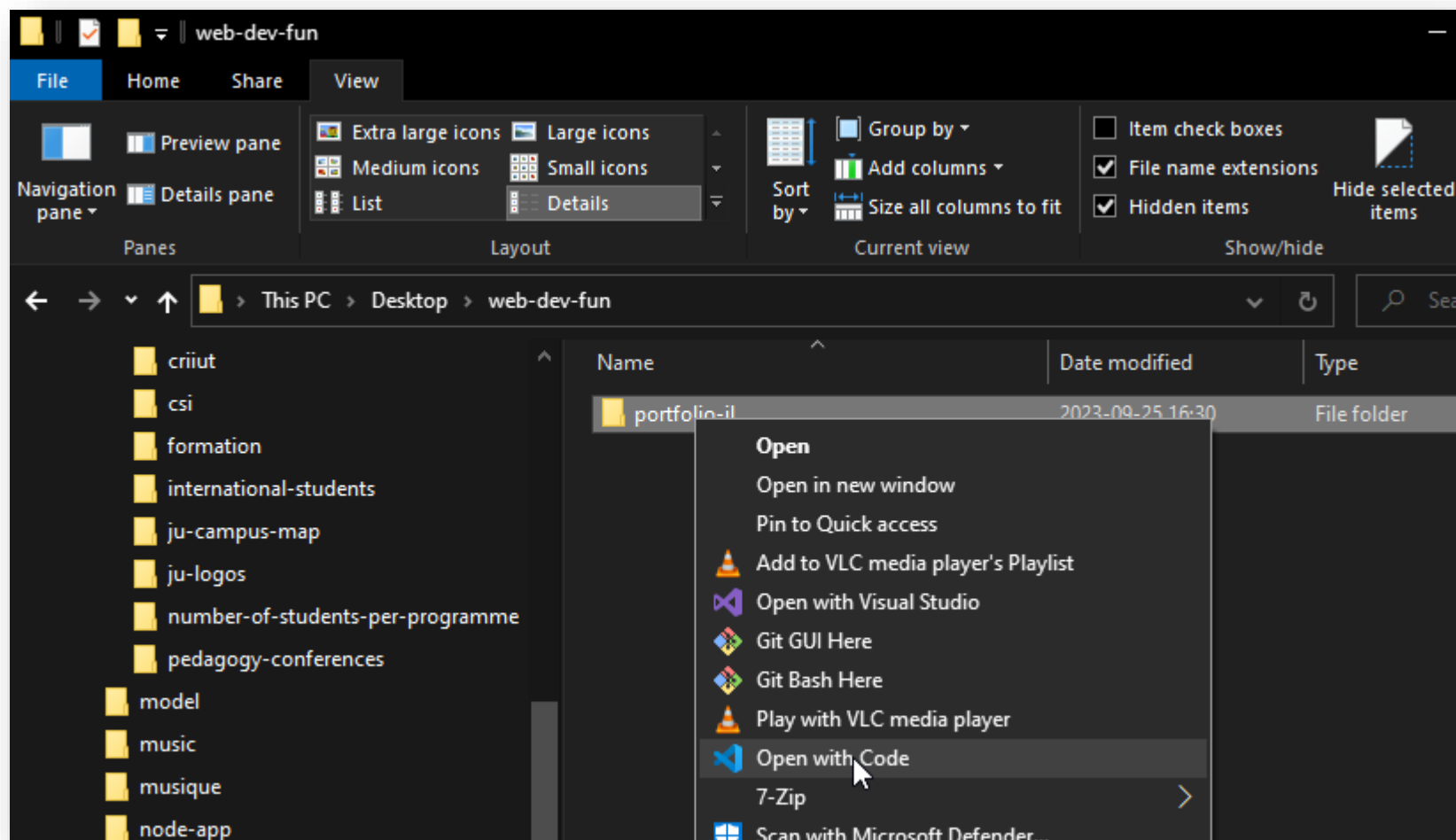
- Copy all your files from your previous project directory to your newly created one



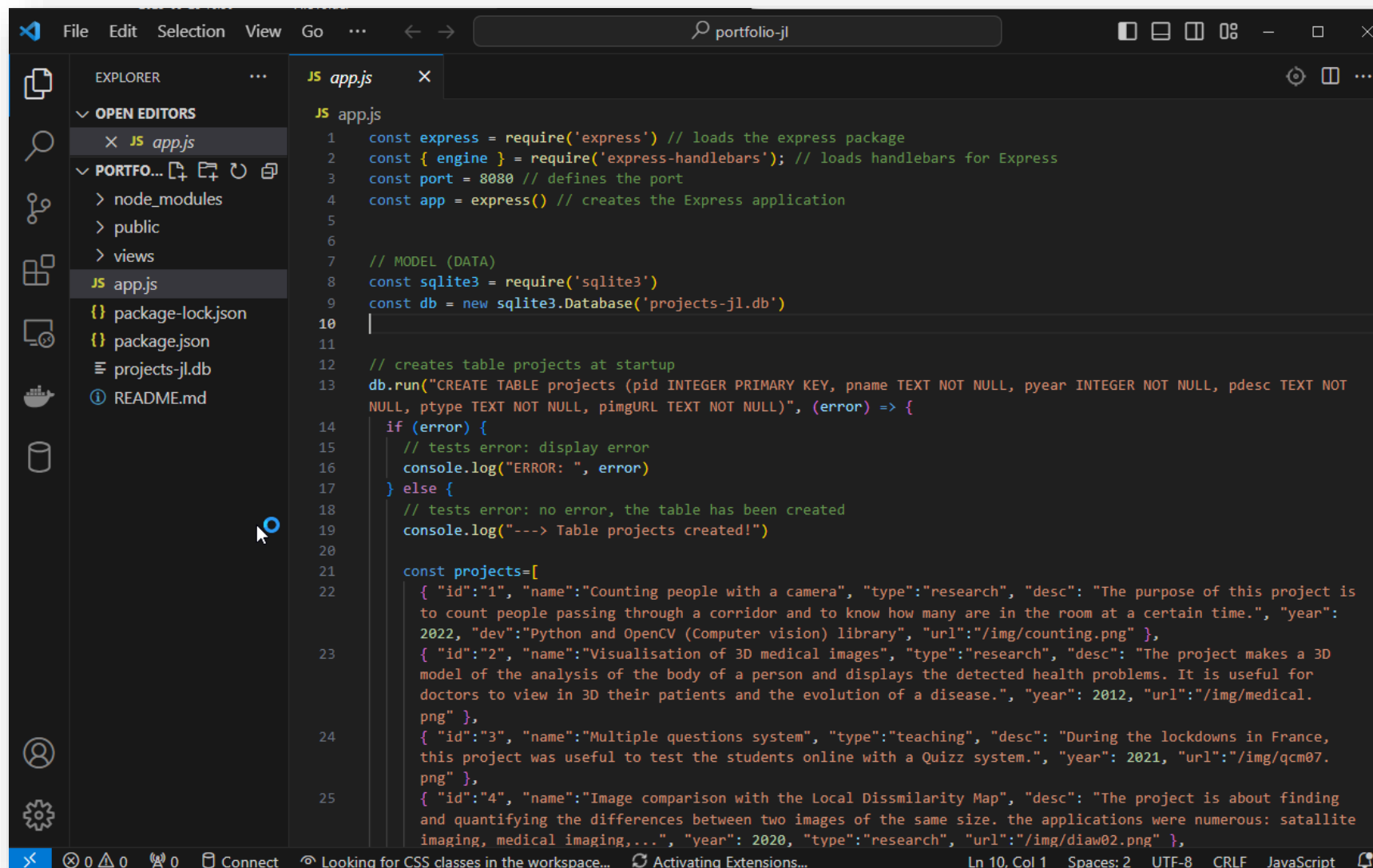
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- We are ready to open the **new working directory** with VSCode:



- We are ready to open the **new working directory** with VSCode:

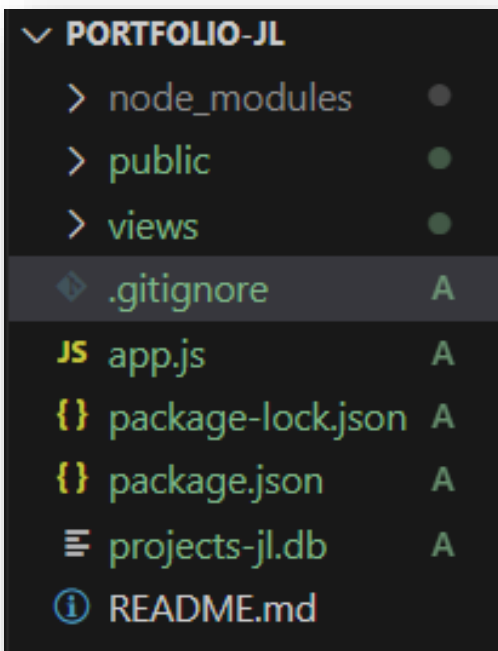


```

1  const express = require('express') // loads the express package
2  const { engine } = require('express-handlebars'); // loads handlebars for Express
3  const port = 8080 // defines the port
4  const app = express() // creates the Express application
5
6
7  // MODEL (DATA)
8  const sqlite3 = require('sqlite3')
9  const db = new sqlite3.Database('projects-jl.db')
10
11
12  // creates table projects at startup
13  db.run("CREATE TABLE projects (pid INTEGER PRIMARY KEY, pname TEXT NOT NULL, pyear INTEGER NOT NULL, pdesc TEXT NOT NULL, ptype TEXT NOT NULL, pimgURL TEXT NOT NULL)", (error) => {
14      if (error) {
15          // tests error: display error
16          console.log("ERROR: ", error)
17      } else {
18          // tests error: no error, the table has been created
19          console.log("---> Table projects created!")
20      }
21
22      const projects=[
23          { "id": "1", "name": "Counting people with a camera", "type": "research", "desc": "The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time.", "year": 2022, "dev": "Python and OpenCV (Computer vision) library", "url": "/img/counting.png" },
24          { "id": "2", "name": "Visualisation of 3D medical images", "type": "research", "desc": "The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease.", "year": 2012, "url": "/img/medical.png" },
25          { "id": "3", "name": "Multiple questions system", "type": "teaching", "desc": "During the lockdowns in France, this project was useful to test the students online with a Quizz system.", "year": 2021, "url": "/img/qcm07.png" },
26          { "id": "4", "name": "Image comparison with the Local Dissimilarity Map", "desc": "The project is about finding and quantifying the differences between two images of the same size. the applications were numerous: satellite imaging, medical imaging,...", "year": 2020, "type": "research", "url": "/img/diaw02.png" },

```

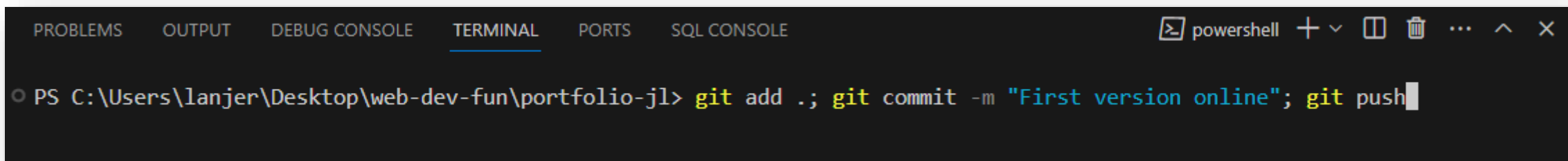
- Now, we want to synchronize our working directory with our GitHub repository
- However, we don't want to store node_modules online (it is not useful and bad for the planet!)
- To avoid sending the node_modules folder, create a file ".gitignore", it will tell git you want to ignore the changes to "node_modules" and do not want to send it online!



```
◆ .gitignore
1  # ignore ALL .log files
2  #*.log
3
4  # ignore ALL files in ANY directory named temp
5  node_modules/
6  |
```

4) ADD, COMMIT & PUSH

- Now, we want to synchronize our working directory with our GitHub repository
- However, we don't want to store node_modules online (it is not useful and bad for the planet!)
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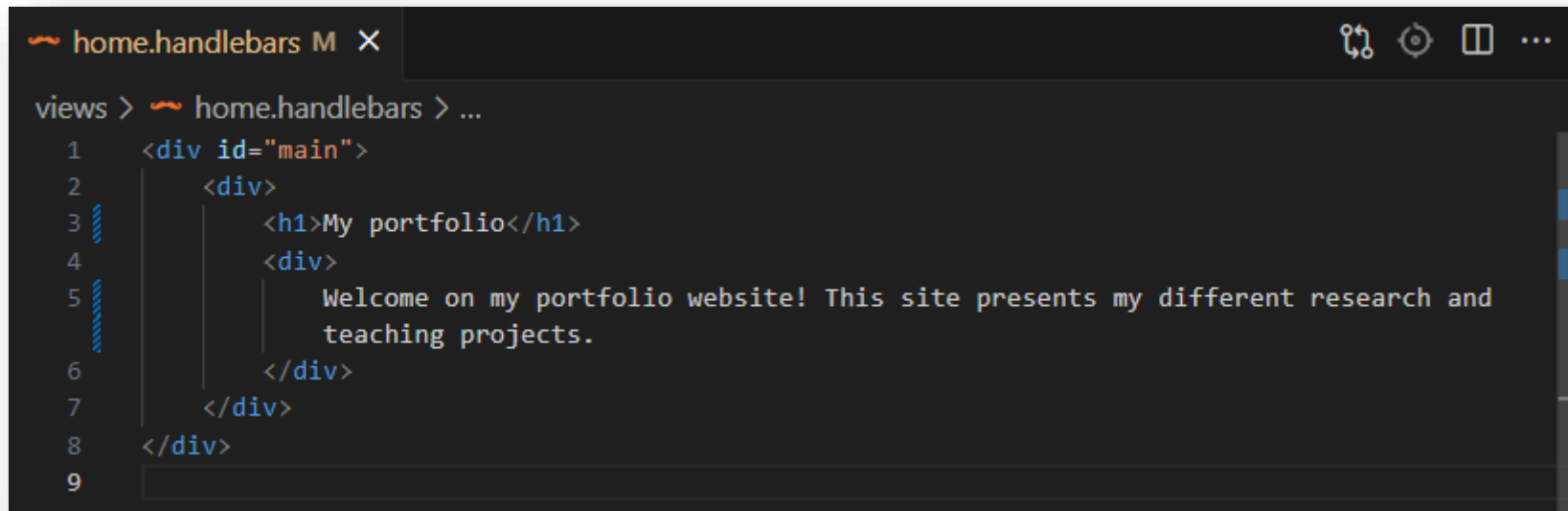
A screenshot of a Visual Studio Code terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (which is active), PORTS, and SQL CONSOLE. The terminal title bar says 'powershell'. The command prompt shows the path 'C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl' and the command 'git add .; git commit -m "First version online"; git push' being entered.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL CONSOLE powershell + - [ ] [ ] ... ^ X  
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> git add .; git commit -m "First version online"; git push
```

```
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> git add .; git commit -m "First version online"; git push
[main 732eec9] First version online
21 files changed, 2896 insertions(+)
create mode 100644 .gitignore
create mode 100644 app.js
create mode 100644 package-lock.json
create mode 100644 package.json
create mode 100644 projects-jl.db
create mode 100644 public/css/spectre-0.5.9.css
create mode 100644 public/css/styles-jl.css
create mode 100644 public/img/bandeau-html-css.png
create mode 100644 public/img/counting.png
create mode 100644 public/img/diaw02.png
create mode 100644 public/img/jl.svg
create mode 100644 public/img/management.png
create mode 100644 public/img/medical.png
create mode 100644 public/img/php01.png
create mode 100644 public/img/qcm07.png
create mode 100644 views/404.handlebars
create mode 100644 views/home.handlebars
create mode 100644 views/layouts/main.handlebars
create mode 100644 views/partials/menu.handlebars
create mode 100644 views/project.handlebars
create mode 100644 views/projects.handlebars
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 8 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (29/29), 1.06 MiB | 1.29 MiB/s, done.
Total 29 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/jlandre72/portfolio-jl.git
   db5f590..732eec9  main -> main
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> █
```

4) ADD, COMMIT & PUSH

- If we change any file, we need to add, commit and push again to synchronize the changes with the GitHub server!



```
home.handlebars M X
views > home.handlebars > ...
1 <div id="main">
2   <div>
3     <h1>My portfolio</h1>
4     <div>
5       Welcome on my portfolio website! This site presents my different research and
6       teaching projects.
7     </div>
8   </div>
9 </div>
```

```
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> git add .
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   views/home.handlebars

PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> git commit -m "modified home.handlebars v1.1"; git push
[main 2fabf32] modified home.handlebars v1.1
 1 file changed, 1 insertion(+), 1 deletion(-)
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 357 bytes | 357.00 KiB/s, done.
Total 4 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/jlandre72/portfolio-jl.git
   b981351..2fabf32  main -> main
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> █
```

 main  1 branch  0 tags

Go to file

Add file ▾

<> Code ▾

 jlandre72 modified home.handlebars v1.1

2fabf32 now  4 commits

 public	First version online	47 minutes ago
 views	modified home.handlebars v1.1	now
 .gitignore	First version online	47 minutes ago
 README.md	Initial commit	1 hour ago
 app.js	First version online	47 minutes ago
 package-lock.json	First version online	47 minutes ago
 package.json	First version online	47 minutes ago
 projects-jl.db	modified home.handlebars v1.0	1 minute ago

 jlandre72 modified home.handlebars v1.1

2fabf32 · 1 minute ago

 History

Name	Last commit message	Last commit date
 ..		
 layouts	First version online	48 minutes ago
 partials	First version online	48 minutes ago
 404.handlebars	First version online	48 minutes ago
 home.handlebars	modified home.handlebars v1.1	1 minute ago
 project.handlebars	First version online	48 minutes ago
 projects.handlebars	First version online	48 minutes ago

ANY QUESTIONS?